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Education

- **California State University, Fullerton** – MS in Computer Engineering

- **University of California, Irvine** – BS in Software Engineering
  - Completed with a GPA of 3.671.
  - For a year, I participated at ACM club meetings, where I:
    - Verbalized technical problem-solving approaches for *easy* and *medium* LeetCode problems to an audience of various skill levels.

Listed Skills

- Python
- C++
- Rust
- JavaScript
- HTML5; CSS3
- PHP
- Roblox Lua(u)
- x86 Assembly

- MongoDB
- MySQL
- Next.js; React
- Docker Compose
- Git; GitHub
- GNU/Linux
- Bash
- PowerShell
- Android/Linux
- FFmpeg

- Affinity Photo
- FL Studio
- OBS Studio
- Filmora 13
- VS Code; VSCodium
- Microsoft Office

Employment

- **Engineering Assistant @ Professional Engineering Center** – June 2015 thru present
  - Safely managing high-impact private data going back since 2003, enabling reliable file access for the engineering team.
  - Translating hand-drawn sketches into production-ready AutoCAD drawings for review by a licensed engineer, accelerating project turnaround.
  - Directly contributing to structural designs for high-stakes projects at multi-national manufacturers and regional utility providers.
  - Reverse-engineered and modified legacy plugins to fully support modern versions of AutoCAD, enabling full migration to an updated toolchain and reducing cognitive demand upon transition.

- **Don't Wake the Labubus @ INSIDR GAMES** – Sep thru Nov 2025
  - https://www.robloxgo.com/game/84920227080693/Dont-Wake-The-Labubus
  - Contributed to a Roblox game reaching 369,800 plays and generating a total of ~100,000 robux in revenue.
  - Integrated core-loop mechanics that helped the game achieve an 87% rating.
  - Diagnosed and fixed difficult-to-triage bugs across existing gameplay systems.
  - Participated in developer discussions on power-up balancing, thresholds, and selection of third-party assets.

- **Raising Cane's and El Pollo Loco** – Aug 2019 thru June 2021
  - Communicated with customers and team members in a fast-paced environment.
  - Operated drive-thru and dine-in cash registers with total revenue exceeding $3000 per day; processing orders through the POS system and supporting overall restaurant operations.
    - Helped maintain steady order throughput during peak hours.

On-call Designer @ Islamic Institute of Orange County et al. – April 2016 thru present

- Controlling graphic design for printed and digital materials for social events leveraging Affinity Photo and Canva, ensuring brand consistency and on-time delivery.
- Creating marketing copy for affiliated non-profits such as Sabil USA, BSA Scouts, and Girl Scouts of America.
- Matching design language with existing literature and coördinating with print shops, reducing revision.
- Additionally, designed real-estate visuals for agents contracted by Berkshire Hathaway.

Research Internship @ CSU Fullerton – Sep 2017 thru May 2018

- Analysed *my first* proper code base and documented findings that helped the research team streamline future development.
- Supported graduate students in building a Java-based brain-computer interface targeting users with limited mobility.
- Coordinated with Professor Kiran George under the Val Tech Program to log internship hours, ensuring that I and the team met our reporting requirements.
- *My assigned task:* To design a mock user interface for initial calibration procedure.

- **Salient Moment:** I identified and took the initiative to solve issues which were outside my purview.
- During a meeting, I overheard that the researchers were facing challenges in accessing a specific data structure obscured by dependency code, which blocked their analysis.
- I located the data structure in a `private` variable buried in a library class.
- To add some spice, I drafted a digital illustration showing the variable's name and presented that to the group.
- However, my solution was not actionable as it required changes to the dependency code and overlooked other project needs.

Projects & Research

Roblox Freedom Distribution – July 2023 thru present

- https://github.com/Windows81/Roblox-Freedom-Distribution
- I have been building games on Roblox from 2013, having had an account since 2008.
- At the time, the company was a small Silicon-Valley start-up.
- Over the years, its developer-friendly tools led me to take up my current-day career as a developer.
- From 2023, I've built a scalable application for users to *host* and *join* self-hosted, historic versions of Roblox in a local- or wide-area network.
- Created a Python bootstrapper, dually for end users and custom backends, to enable seamless connectivity.
- Ensured stable standard of quality through team collaboration, x86-assembly patches, and lots of technical code review.
- Consulted a legacy C++ codebase to extrapolate compiled-program behaviour.
- Built a automation asset-processing pipeline for proprietary 3-dimensional mesh and rbxl formats, promoting product quality.
- Evaluated fair-use compliance by auditing business-legal literature, publicly disclosing some analysis of legal risks and benefits of various intended use cases.

Frontend Developer at GameIn – Sep 2023 thru Mar 2024

- Before graduating from UC Irvine, I took a capstone course where I collaborated, among classmates, with one of a choice of several companies.
- Owing to my prior experience in video-game development, I chose to work with GameIn, who were building a tool to match e-sports streamers and paying sponsors.
- Implemented mobile-user-friendly pages; leveraged Next.js (and thus React) from mockups that we also made.
- Arranged, as part of a team, a slide deck with stylistic help from Figma.
- The slide deck was presented to an audience of approx. 50 other students and their families.

Bring Back Buffets – March & April 2021

- https://github.com/Windows81/Bring-Back-Buffets
- During the Covid lockdowns, I built a mobile web app to revive buffets whilst minimising infection risk.
- This project's user-interaction model drew inspiration from how some Korean barbecue establishments operate:
- Where dine-in customers can order food directly from a website they can access from their phones.
- I built separate user interfaces for customers, cashiers, and back-of-house staff, with step-by-step documentation for setup.

GoAnimate Wrapper – Dec 2019 thru Nov 2020

https://github.com/GoAnimate-Wrapper/GoAnimate-Wrapper

- Using GoAnimate's *Legacy Video Maker*, I animated some engaging videos on my YouTube channel, often satirising trends set forth by other publishers who use the same tool.
  - However, it needed a remote server for file persistence and access to crucial assets, which was being discontinued after 2019.
  - A newer version of the program existed, but lacked crucial animation assets and effects.
- GoAnimate Wrapper is a server-backend application designed to interop with Vyond's Legacy Video Maker.
  - Reverse-engineered Flash files and analysed legacy ActionScript code.
  - Created and scaled a modern JavaScript interface replicating GoAnimate's back-end functionalities.
- With a team of experts managing the community up to this day, Wrapper became my most popular application.
- Because of the need that it filled, my tool delivered hundreds of engaging video experiences.
  - By May 2021, over 21,000 results would appear on a YouTube query for Wrapper and its successors.

Responsible Security Disclosure - YuJa API – Dec 2022

https://github.com/Windows81/Screwdja-YuJa

- I identified and responsibly disclosed a high-impact vulnerability in YuJa's API.
- Over 6,000,000 academic videos, including private recordings of students taking proctored exams, were exposed on the open net.
- YuJa released a security patch after ~2 days, closing the loop-hole for good.

Webhooky – Feb 2018 thru Sep 2019

https://github.com/Windows81/Playing-Webhooky

- Discord's webhook API allows developers to easily send custom messages from live Roblox games to a shared space on Discord, where data persistence is more accessible than on Roblox's servers.
- In 2018, when Discord abruptly blocked all webhook activity from Roblox servers, I built a lightweight proxy to suit that need.
  - Made use of PHP, SQL (MySQL), and PHPMyAdmin to process and store request logs.
- Served over 200 distinct games, including from major studios such as ROLVe.
- Monitored and tracked infrastructure usage stats on cPanel via 000webhost, avoiding excessive downtime.

Tubeup – Aug 2023

- I use a variety of tools, most of which I install from GitHub, to automate tasks that would otherwise take me months to complete.
  - However, some of these tools are missing features or contain bugs which are trivial to fix.

https://github.com/bibanon/tubeup

- Tubeup is a Python utility with 478 stars and new versions being released as of Feburary 2026.
  - It uses the `yt-dlp` and `internetarchive` libraries to download videos from YouTube to re-upload to the Internet Archive.
- I discovered Tubeup in 2023 to help mitigate the risk of my prior work on YouTube being undiscoverable due to shifting policies.
- On a batch of 234 videos, the program crashed on attempting to download deleted or invalid media.
- I patched Tubeup to alleviate the issue, then I submitted my changes via a pull request on the official project page.
- Unlike most other projects to which I contributed, the author quickly accepted my changes.